

Final Assignment – Generative AI Systems Design & Implementation

Student 1	Your Full Name	Your email
Student 2	Your Full Name	Your email

Overview

This final assignment is designed to consolidate your learning in Generative AI systems design and implementation. You will select a relevant business problem, conduct market and technical research, design an AI-driven system architecture, implement your solution, and present it to your peers. The process is intended to simulate a real-world product development lifecycle, integrating both technical skills and business thinking.

Assignment Steps (*percentage out of total grade*):

1. Select a business problem (preferably one you know from your work or daily life). (5%)
2. Conduct deep market research and technical discovery with relevant stakeholders.(15%)
3. Create a Generative AI-based system architecture to solve the chosen problem.(20%)
4. Implement the solution. (50%)
5. Pitch your solution to the class, highlighting both technical achievements and business impact.(10%)

Before starting implementation, review the Project Breakdown slides shared in class. They define the working methodology - especially how to scope your POC before writing any code.

1. Problem Selection & Definition (5%)

Chosen Business Problem

Provide a clear and concise statement of the business problem you have selected. This should be specific, measurable, and relevant to your experience or industry.

Example: *In my role as a [position] at [company/organization], I have observed [problem description]. This problem impacts [stakeholders] by [quantify impact if possible, e.g., time lost, revenue leakage, customer dissatisfaction].*

Background & Context

- Industry/Domain:
- Current Processes or Systems:
- Pain Points:
- Business Impact:

2. Market Research & Technical Discovery (15%)

In this section you will perform all the prep work necessary to help you direct the system you will be building to solve your chosen problem.

Market Landscape

- **Existing Solutions:** Identify and evaluate current tools or competitors.
- **Market Gaps:** Highlight where these solutions fall short.
- **Target Audience:** Define who will use your solution and why they need it.

Technical Discovery

- **Stakeholder Interviews:** Summarize insights from relevant team members, domain experts, or end-users.
- **Data Availability & Quality:** Identify datasets, APIs, or system integrations available
- **Feasibility Assessment:** Technical constraints, costs, and operational considerations.

3. Proposed GenAI System Architecture (20%)

In this section you will be describing your proposed solution to the problem, down to the

smallest details. Your architecture should take into account your conclusions from the previous section.

High-Level Overview

- **Solution Concept:** Briefly describe how your GenAI-based solution will address the problem.
- **Key Functionalities:** Outline core features and their benefits (technical benefits as well as business benefits) .

System Architecture Diagram

Provide a clear diagram illustrating:

- Data sources
- Preprocessing pipelines
- GenAI components
- Output delivery & feedback loops

(You can use tools like Lucidchart, Figma, draw.io, or PowerPoint to make this visual.)

Technology Stack

Fill the following table with your tech stack

Component	Technology Choice	Reason
Technological component (e.g. LLM)	Your chosen product/service (e.g. OpenAI GPT-4o)	Explanation for your choice (e.g. <i>High accuracy and contextual reasoning</i>)

4. Implementation(50%)

In this section you are expected to implement the architecture you have described. Please describe your implementation process, challenges you encountered and how you overcame them, and the performance of your solution.

Scoping Your POC

Before building, define:

- Input: What does the system receive?
- Output: What does it produce?
- Success metric: How will you know it works?
- Minimum viable test set: At least [X] examples you'll evaluate against
- Target: Reach [X]% on your metric before adding complexity

Start with the strongest available model and the simplest possible prompt. Only optimize after you have a baseline.

Development Steps

1. **Data Preparation** – [What you did to prepare/clean/enrich data]
2. **Model Integration** – [How you connected LLM APIs or fine-tuned models]
3. **Application Logic** – [Business logic and rules implemented]
4. **Testing & Validation** – [Test cases, evaluation metrics, user feedback]

Challenges & Solutions

- **Challenge:**
- **Solution:**
- **What if your POC didn't hit the target metric?** Describe what you tried, what you learned, and how you adjusted scope or approach. A well-documented failure is worth more than a fudged success.

System Performance (Technical & Business Impact)

Technical KPIs

Metric	Description	Target	Achieved
Accuracy	% of correct/appropriate responses	90%	[Value]
Latency	Avg. time from user input to response	< 2 sec	[Value]

Uptime	System availability	99%	[Value]
Error Rate	% of failed or incorrect responses	< 5%	[Value]

Business KPIs

****NOTE:** for the following section you can use automatic evaluation metrics (e.g for productivity you can measure how long it takes your system to do a task compared to the estimated time it takes a human)

Metric	Description	Baseline	After Implementation	Improvement
--------	-------------	----------	----------------------	-------------

Cost Reduction	Operational cost savings from automation	[\$X]	[\$Y]	[Z]%
Productivity Gain	Increase in tasks completed per employee	[X/hr]	[Y/hr]	[Z]%
Customer Satisfaction	Average feedback rating from users/customers	x/5	(x+0.8)/5	+0.8
Revenue Impact	Additional revenue attributed to system	[\$X]	[\$Y]	+[Z]%

Business Impact Highlights

- **Operational Efficiency:** Reduced manual processing time by [X%], enabling staff to focus on high-value tasks.
- **Cost Savings:** Estimated annual savings of \$[amount] due to automation and reduced error correction.
- **Customer Retention:** Faster, more accurate responses improved client trust, reducing churn by [X%].

- **Market Advantage:** System capabilities positioned the business as an early adopter of GenAI, enhancing competitive differentiation.

Strategic Scalability Potential

- Ability to integrate into additional departments with minimal extra cost.
- Potential to extend to new markets or customer segments.
- Predictive analytics integration could further enhance decision-making.

Screenshots / Code Snippets

(Attach relevant visuals, UI screenshots, or sample API responses.)

Code repository

A link to your full code repository on github.

5. Pitch to Class(10%)

You will be pitching your system/solution/product to the class, do your best! Below is a flow proposal, but it is entirely up to you. Remember clarity and simplicity are key.

- **Hook:** Open with a compelling fact or story.
- **Problem:** explaining your chosen problem (including why you chose it)
- **Solution Overview:** Summarize in plain language.
- **Demo:** Quick walkthrough.
- **Business Value:** Quantify ROI.
- **Call to Action:** Suggest next steps.

Appendices

****FULL CODE (a github repo link)****

GOODLUCK!!